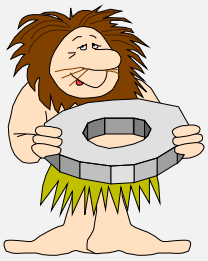
INTELLECTUAL PROPERTY

- **Patents**
- **Copyrights**
- **Trademarks**
- **Trade Secrets**



EAS5450 Class #4

1

Patent Protection is a Contract with Society

To promote "progress of science and the useful arts..."

- Stimulates innovation by rewarding inventors with a limited (15-20 year) exclusive right.
- Encourages public disclosure of inventions rather than maintaining them in secrecy:
  - Must file within 1 year of public disclosure.

2

RIGHTS FLOWING FROM PATENTS

- Right to exclude others from making, using, selling, offering to sell, or importing
- Legal Remedies:
  - Injunction
  - Monetary damages for lost profits

3

STRATEGIC REASONS FOR PATENTS

- **Create Barriers to Entry**
- **Improve Bargaining Position/Trading**
  - Venture Capital Value
  - Strategic Alliance Value
- **Corporate Image**
- **Royalties**
- **Innocent Infringement is Actionable**

"currency" for doing deals

4

PATENT DISADVANTAGES

- Disclosure
- Expiration
- Time & Cost to File, Prosecute & Renew
- Time & Cost to Enforce
- Value is Uncertain Before Filing

5

ASSIGNMENTS & LICENSING

IP can be bought and sold:

Patents can be assigned;
Patents can be bought/sold outright.

IP can be licensed:

License can be non-exclusive;
License can be completely or partially exclusive (e.g., limited to certain markets, products or geographical region).

6

LICENSING: FACTORS TO CONSIDER

- **Resource availability:**
  - Product development
  - Production
  - Marketing
  - Capital

7

ADVANTAGES OF LICENSING

- Rapid Market Diffusion
- Rise to Dominant Design
  - Become Technology Standard
- Attract 3rd-Party Developers
- Achieve Economies of Scale (Pricing)

8

WHOLLY PROPRIETARY vs. WHOLLY OPEN

Wholly Proprietary	Limited Licensing	Moderate Licensing	Liberal Licensing	Wholly Open
Xbox Console	Xbox Games	MSFT Windows	Sun's Java	LINUX Oper Sys

From: Schilling, M. "Strategic Management of Technological Innovation." McGraw-Hill 2008

9

Fraunhofer IIS : MP3 Technology

Liberal licensing led to Dominant Design in audio compression format

10

TECH + HEALTH 06.15.14

Tesla's Radical Patent Move is a Plot to Take Over the Road

The founder of the world's sexiest electric car company announced that he's opening up patented Tesla technology to anyone who wants it. But it only looks like philanthropy: what Tesla really wants is for other car companies to be more like them.

Tesla founder Elon Musk shocked the world with his latest convention-wisdom defying move this week. In a blog post, he announced that the company would open up its patents and other intellectual property to competitors. "Tesla will not initiate patent lawsuits against anyone who, in good faith, wants to use our technology," he wrote.

11

To Be Patentable, An Invention Must Be …

1. Statutory Subject Matter
2. Novel
3. Non-Obvious
4. Useful (Utility Patents)

12

1. STATUTORY SUBJECT MATTER

Patentable:

- Machine
- Manufacture
- Composition of Matter
- Process
- Improvements

Not Patentable:

- Abstract Ideas
- Laws of Nature
- Naturally Occurring Phenomenon
- Mathematical Algorithms
- Humans

13

2. NOVELTY

- **The invention has not been:**
  - in public use/on sale/described in a printed publication > 1 year; or,
  - the subject of a US patent with a filing date before your invention date.

14

3. NON-OBVIOUSNESS

The invention must be:

- non-obvious
- to a person having ordinary skill in the art
- to which subject matter pertains
- at the time the invention was made

15

4. USEFUL

The invention must be:

- useful or functional for a utility patent
- this is a low bar usually
- does not apply to design patents

16

Patentable…

- **Adaptations of known technologies:**
  - improvements to known technologies
  - combinations of known technologies
  - new uses of known technologies

17

TYPES OF PATENTS

- **Utility Patents**
  - Processes, devices, inventions
  - New, useful, non-obvious
  - Term: 20 years from date of filing
- **Design Patents**
  - Non-functional features of useful objects
  - Ornamental, new, non-obvious, inseparable
  - Term: 15 years from date of grant
- **Plant Patents**
  - New varieties asexually reproducible
  - New, non-obvious
  - Term: 20 years from date of filing

18

SOFTWARE PATENTS

Computer programs, per se, are not statutory subject matter; but practical applications of them are statutory subject matter:

- auto spell checker routines,
- video graphics routines,
- encryption methods.

Software is patented and/or copyrighted or neither.

19

Supreme Court Overturns Software Patents…

- July 2010: *In re Bilski*
  - US Supreme Court rejects patent on a method of hedging risks in commodities trading.
- June 2014: *Alice v. CLS Bank*
  - Rejects "do it on the computer" business method patent.
  - Rejection rate of these software patents rose from 24% to 78% after ruling.

20

Pharmaceutical Patents

- **20 year patent**
  - R&D: 1 in 5,000 make it to market
  - 1-12 years: New Drug Application Process
  - 12-20: MARKET OPPORTUNITY
  - 20-?: Generic drug entries; revenue plummets
- **Pharmas use IP to extend protection**
  - "Hard Switch" patent strategy*

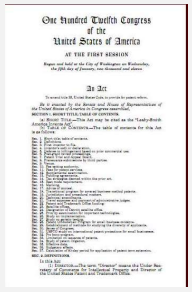
**Bloomberg Businessweek, "Drugmakers' Patent Tactic Faces a Big Test" (Apr 13-19, 2015)*

21

AMERICA INVENTS ACT

Signed into law: Sept 16, 2011

- **First-Inventor-to-File**
  - Effective Mar 16, 2013
  - Protection against derivation by prior applicant.
  - 1-year grace period retained
    - "First to Publish"
    - "First to Sell"
    - "First to File Provisional Patent"

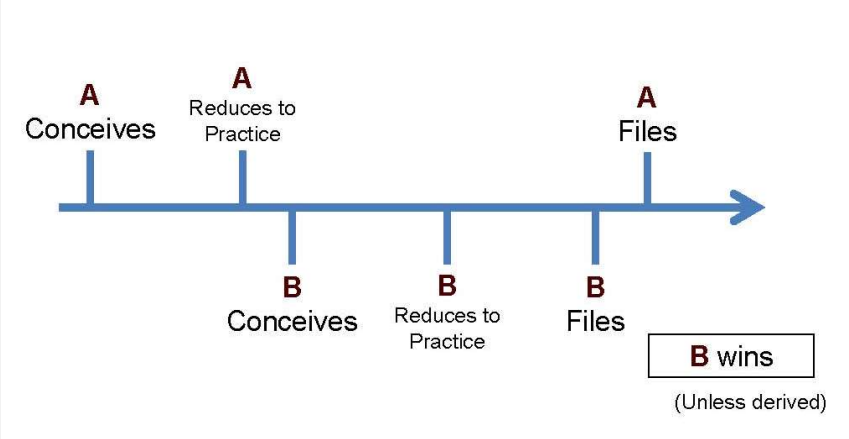


One hundred Twelfth Congress
of the
United States of America
AT THE FIRST SESSION

An Act

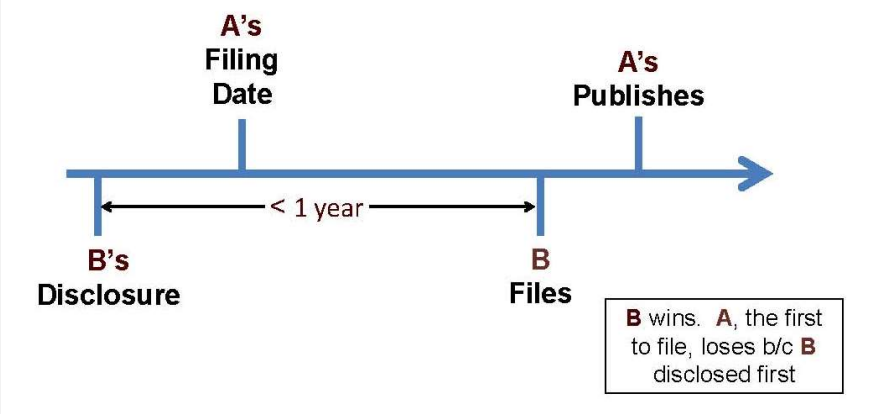
22

First-to-File



23

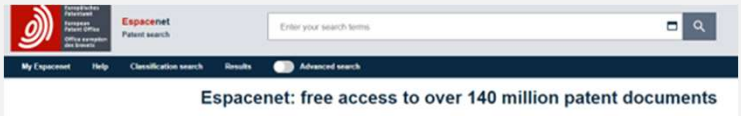
First-to-Publish



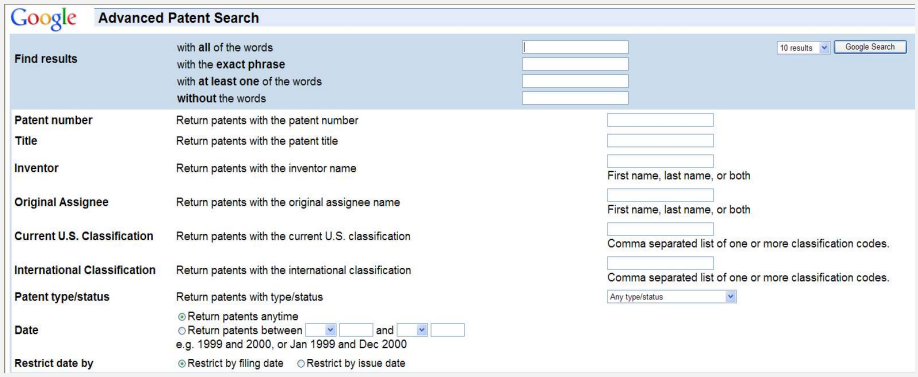
24

PATENT APPLICATION PROCESS

- **Keep careful records**
  - Document all efforts
  - Date and sign with witness
- **Do a patent search**
  - Patent attorney
  - www.uspto.gov
  - www.google.com/advanced_patent_search
  - www.nolo.com
  - Derwent Innovations Index (www.library.upenn.edu)
  - www.espacenet.com



25



26

PATENT APPLICATION PROCESS

- **Preparation:**
  - Provisional
  - Utility
- **Filing**
- **Examination (Prosecution: 32 wk avg)**
- **Publication**
- **Post-Grant Review**

27

INTERNATIONAL PATENTS

-- Protect where product MADE or SOLD --

- **International Treaties**
  - Paris Convention
    - World Intellectual Property Organization
  - Patent Cooperation Treaty
- **European Patent Office**

28

STARTUPS WITH PATENTS ARE MORE VALUABLE

- More funding*
- Earlier financing*
- More prominent investors*
- Successful IPOs†
  - 31.5% with patents vs 7.25% without.
- Lower bankruptcy rate†
  - 5.6% with patents vs 14.2% without.

**Hsu, D. & R. Ziedonis, "Patents as Quality Signals for Entrepreneurial Ventures" (2007)*
† Coo, J. & P. Hsu, "The Informational Role of Patents in V.C. Financing" (6/8/2011)

29

STARTUP PATENT TRENDS

- 1/3 of V.C. funded startups have filed.
- 19% filed at least one application prior to any funding.
- 56% of Kleiner Perkins portfolio companies have filed (typ. of large VCs).
- Trend: since 2005, average startup less likely to apply than prior year (notably software).

Techcrunch.com – 6/21/2012

30

FUNDED COMPANIES WITH PATENTS

INDUSTRY	PATENT RATE
Semiconductor	65.2%
Biotech	61.6%
Hardware	52.0%
Cleantech	47.7%
Mobile	32.0%
Software	29.5%
Enterprise	25.2%
Advertising	23.7%
Ecommerce	10.5%

Techcrunch.com – 6/21/2012

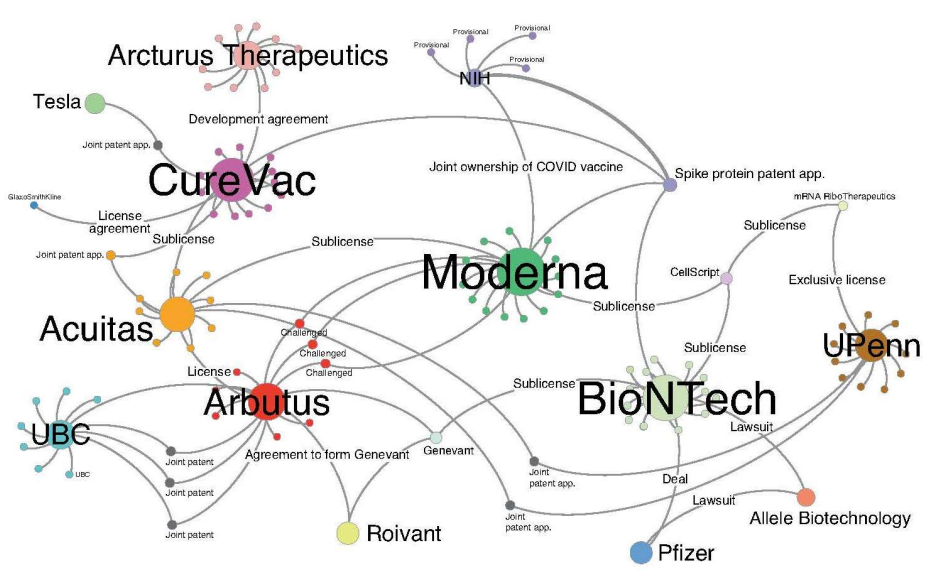
31

2008 survey of biotech startups cites patents as the most important means of capturing competitive advantage, ahead of secrecy and lead time.

2008 Berkeley Patent Survey (6/30/2009)

32

Patent Network of mRNA-Based COVID-19 Vaccine Candidates



Nature Biotechnology – Vol 39 – May 2021

33

HEALTH

Moderna Sues Pfizer and BioNTech, Alleging mRNA Patent Infringement

Moderna says rivals copied its methods for encoding genetic material and used the same chemical modification in their Covid-19 vaccine

By Joseph Walker Follow

Updated Aug. 26, 2022 4:31 pm ET

Moderna Inc. sued rival Covid-19 vaccine makers Pfizer Inc. and BioNTech SE, alleging that the companies' shot infringes on key intellectual property owned by Moderna.

Pfizer and BioNTech's vaccine relies on messenger RNA technology that Moderna had developed and patented years ago, including a chemical modification that avoids prompting an undesirable immune response, Moderna said in lawsuits filed Friday.

Wall Street Journal: Aug 26, 2022

34

COPYRIGHT PROTECTION

- Protect literary, musical, dramatic and artistic works fixed in a tangible medium.
- Author has sole right to print, copy, sell or distribute.
- Infringement occurs only when there has been **access** to the copyrighted work, **and** the accused work is a direct copy or is **substantially similar** in expression.
- No remedy for independent creation of copyrighted work.
- Term: 70 years beyond author's death.
- Work for hire: 95 years from creation.

35

Copyright vests to the creator upon creation

… whether registered or not.

36

COPYRIGHT ACTIONS

- **Register with Copyright Office**
  - Not required, but recommended
- **Place notice on works**
  - © 2012 John Doe
  - Copyright 2012 ABC Corp.
  - Not required, but recommended
- **Cease and desist letters**
- **Litigation**
  - Statutory damages if registered

37

Software Copyright Risk

- Copyrighted Basic Input/Output System (BIOS)

- Reverse engineered by documenting every IBM BIOS output function in response to given input commands.
- List of commands and corresponding functions given to Compaq programmers.
- Without access to IBM code, Compaq programmers wrote code to create identical functionality.

38

TRADEMARK PROTECTION

® vs TM

39

Trademark law protects, “a word, name, symbol or design that identifies the source of a product or service”

- …includes colors, sounds, scents

Trademarks identify brand and perception of quality.

40

The Power of Trademark Branding

41

- Registration provides proof of prior right in legal proceedings.
- Registration ® provides visible notice and deterrent to potential infringers.
- ™ can be used if the mark is in the process of being registered but not yet approved; however, it does not provide the legal protection afforded by registration.
- Registration allows the owner to license the mark to another party.
- Registered marks will likely be disclosed in a search by other parties to avoid infringement.
- In some “first to file” countries, registration is required to enforce rights. In other countries (including the U.S. and Canada), first use of the mark establishes protectable rights, even if not registered.
- Different parties can use the same mark if their products are easily distinguishable and there is no chance of consumer confusion.

42

TRADEMARKS

- **Terms and notice**
  - User is owner of mark
  - Perpetual but must keep up with maintenance fees
  - Product™ → Product®
    Service℠ → Service®
- **Guidelines**
  - Keep the trademark short
  - Avoid common prefixes and suffixes in the industry
  - Understand the meaning of the mark in other languages

43

Generic Trademarks Lose Protection

Aspirin, cellophane, zipper and escalator are former trademarks that owners failed to protect from common usage.

44

Proper Uses of Trademarks

- Trademarks are adjectives and are always followed by a noun:
  - XEROX® copiers
  - PEPSI® cola
- Never use trademarks as verbs.
- Trademarks should be identified as such:
  - ® should only be used for registered marks.
  - ™ can be used for unregistered marks.

45

TRADE SECRET PROTECTION

Information used in one's business which can provide an advantage over competitors: methods, formulas, customer lists, proprietary processes, competitive intelligence.

- **Trade secrets must:**
  - Be well defined.
  - Provide economic value to the company.
  - Be subject to “reasonable measures” to keep them secret.

46

Trade Secrets

- Must be not generally known and must take measures to maintain secrecy.
- No disclosure required.
- Protection does not expire.
- “Reverse Engineering” is legitimate, but misappropriation is actionable.
- “Nebulous” legal protection.

47

Tesla sues former employee for allegedly stealing software code

PUBLISHED FRI, JAN 22 2021•9:08 PM EST UPDATED FRI, JAN 22 2021•9:50 PM EST

Lora Kolodny
@LORAKOLODNY

TESLA

According to a legal filing on Friday, Tesla is suing a former employee and software engineer named Alex Khatilov alleging trade secret theft and breach of contract.

In the complaint, the company accuses Khatilov of grabbing code and files from WARP Drive, a back-end software system that Tesla developed to automate a range of business processes involved in manufacturing and selling its cars. They also accuse him of deleting possible evidence when security teams confronted him.

The company also says it has spent an estimated “200 man-years of work” to develop the code in question.

CNBC 1/22/2021

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Protecting Trade Secrets

- **Non-Disclosure Agreements**
  - Cannot use information for personal benefit.
  - Cannot disclose information to third parties.
  - Not a substitute for patents
    - Does not stop 1-year clock for filing
- **Non-Compete Covenants**
  - Protects against competition from former employees
  - Nebulous legal precedent:
    - Varies from state to state
    - Courts have imposed non-competes where they did not exist, and have struck them down where they did exist.

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WALTER KIDDE PORTABLE EQUIPMENT INC.
394 South Third Street
Mebane, NC 27302

June 8, 1999

X-IT Products, LLC
1435 Bay St., Suite 31
San Francisco, CA 94123

Dear Mr. Ive:

We have recently begun discussions concerning a potential transaction between Walter Kidde Portable Equipment Inc. ("Buyer") and X-IT Products, LLC ("Seller"). In the course of such discussions, Seller may furnish Buyer with certain information concerning the business, financial condition, operations, assets and liabilities of Seller or that may otherwise be "confidential" or "proprietary" (collectively, "Confidential Information"). Both parties wish to confirm their agreement concerning certain restrictions Buyer has agreed to accept as a condition to Seller's disclosure to Buyer of Confidential Information.

Buyer agrees that at all times, (i) it and its representatives (including, without limitation, its directors, officers, employees, agents or advisors) will use the Confidential Information only for the purposes of evaluating the potential transaction with Seller; (ii) it will provide the Confidential Information only to its employees involved in the potential transaction with Seller or outside advisors retained by Buyer in connection with such potential transaction and who agree to keep such information confidential; (iii) it will not disclose, in whole or in part, any Confidential Information (except to the extent Seller gives its prior written consent or other than as specifically provided for herein); and (iv) it will promptly return to Seller, upon ~~Buyer's~~ Seller's request, all copies of any Confidential Information provided to Buyer.

Unless and until a definitive agreement between Seller and Buyer has been executed, neither party will have any legal obligation of any kind whatsoever with respect to any such transaction by virtue of this agreement, except as expressly provided herein. Each party has the right, in its sole discretion, to reject any and all proposals made by the other party and to terminate discussions and negotiations at any time.

For purposes of this agreement, "Confidential Information" shall not, in any event, include any information that (i) is in the public domain (other than through a breach of this agreement by Buyer or any of its representatives) at the time of its disclosure to or use by Buyer, (ii) is in Buyer's possession prior to its disclosure by Seller is independently developed by Buyer (provided that the source of such Confidential Information was not bound by a confidentiality agreement with Seller or otherwise subject to any applicable restriction on the use of such information), (iii) is obtained by Buyer from a third party without any applicable restriction on use, including without limitation, a confidentiality agreement with Seller or (iv) is disclosed pursuant to the order or requirement of a court, administrative agency or other governmental body.

Background & purpose

Agreement

Limited scope

Definition(s)

50

Buyer acknowledges that neither Seller nor any of its representatives makes any representation or warranty, express or implied, as to the accuracy or completeness of the Confidential Information. Buyer agrees that neither Seller nor any of its representatives shall have any liability to Buyer, relating to or resulting from the use of the Confidential Information or any errors therein or omissions therefrom. Only those representations and warranties that are made in a final definitive agreement regarding the transaction, when, as and if expected, and subject to such limitations and restrictions as may be specified therein, will have any legal effect.

This agreement shall be binding on and inure to the benefit of all the parties and their respective successors, heirs, assigns and affiliates and shall be construed under the laws of the State of North Carolina.

By (Signature) 6/8/99

Sam A. Solomon
Chief Financial Officer

WALTER KIDDE PORTABLE EQUIPMENT INC.

Accepted and agreed to:

X-IT PRODUCTS, LLC

By Andrew Ive
(Signature)

A. Ive. 6/8/99
Print Signature Date

CEO
Title

No reps or warranties

Legally binding

No expiration term?

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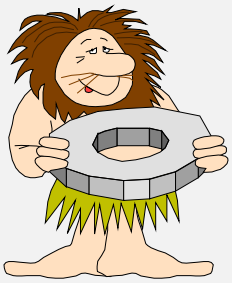
INTELLECTUAL PROPERTY

- **Do we protect it?**
- **How do we protect it?**
- **When do we protect it?**
- **Where do we protect it?**



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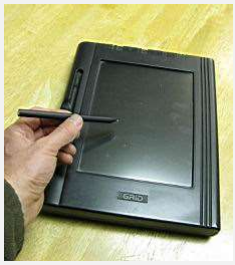
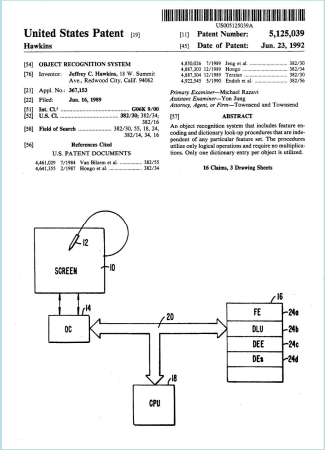


GRiD Pad



United States Patent [19]
Hawkins

[11] Patent Number: 5,125,039
[45] Date of Patent: Jun. 23, 1992

[54] OBJECT RECOGNITION SYSTEM

[76] Inventor: Jeffrey C. Hawkins, 18 W. Summit Ave., Redwood City, Calif. 94062

[21] Appl. No.: 267,153

[22] Filed: Jun. 10, 1989

[51] Int. Cl.[5] G06K 9/00

[52] U.S. Cl. 382/30; 382/34; 382/16

[58] Field of Search 382/30, 55, 18, 24, 382/34, 36, 16

[56] References Cited
U.S. PATENT DOCUMENTS

4,841,029 7/1984 Van Bilzen et al. 382/55
4,841,555 2/1987 Hongo et al. 382/54
4,850,026 7/1989 Jang et al. 382/30
4,857,202 12/1989 Hongo 382/54
4,857,304 12/1989 Terzian 382/90
4,922,545 5/1990 Endoh et al. 382/56

Primary Examiner—Michael Razavi
Assistant Examiner—Yon Jang
Attorney, Agent, or Firm—Townsend and Townsend

[57] ABSTRACT

An object recognition system that includes feature encoding and dictionary look-up procedures that are independent of any particular feature set. The procedures utilize only logical operations and require no multiplications. Only one dictionary entry per object is utilized.

16 Claims, 3 Drawing Sheets

Palm Pilot

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54

Harvard Business School 9-D RA FT
January 19, 1996

Palm Computing, Inc. (B)

Jeff Hawkins poured over his licensing agreement with GRiD. He knew that the *device license* GRiD had sold to Microsoft had been restricted to hardware/software devices they manufactured. It was virtually worthless if the licensee was not a full system producer. He thought that GRiD's recent decision to leave hardware manufacture to others and concentrate on the software side of the business might have the same effect on his 1988 agreement with the company. If he could demonstrate that GRiD's new strategy required his consent, he would be in a much stronger negotiating position. He reexamined Sections 2.B of the licensing agreement

> Mr. Hawkins also grants to GRID a perpetual, irrevocable, exclusive, worldwide license to reproduce, distribute, and sublicense the Program *to end users in* object code form only. GRiD *may not sublicense the Program through resellers* without the prior approval of Mr. Hawkins unless such reseller is also purchasing or licensing GRiD hardware. (Italics added.)

56

Harvard Business School **9-DRAFT**
January 22, 1996

Palm Computing, Inc. (C)

Hawkins hammered out an agreement with Tandy Corporation that expanded the PalmPrint rights he had granted to GRiD in the 1988 license. Hawkins agreed to grant GRID a "nonexclusive, perpetual, irrevocable, worldwide license under all Intellectual Property" that carried the following new rights:

> 2.1.5 to sublicense the Program and derivative works thereof in machine-readable, source (subject to Section 12.1.6 below) and object code form through resellers if such Program or derivative work is bundled with and imbedded in other GRiD or Tandy Corporation *or their Affiliates'* applications or operating systems software. The Program and derivative works thereof may not be sublicensed unbundled to other software or hardware manufacturing companies.
>
> 2.1.6 to utilize source code of the Program and derivative works thereof pursuant to Sections 2.1.3, 2.1.4, and 2.1.5 hereunder if such source code is bundled with and imbedded in GRiD or Tandy Corporation *or their Affiliates'* applications or operating systems software; provided, however, that any such transfer or license of source code be conditioned upon the execution of a confidentiality agreement and upon the giving of notice to HAWKINS 15 days prior to any such transfer or license. Such notice shall identify the party to whom the source code will be transferred, shall specify the source code being so transferred or licenses, and shall include a copy of the confidentiality agreement to be entered into.

In return for extending the license, GRiD granted to Hawkins a royalty-free nonexclusive worldwide license:

> 6.1.1 to use the source and object code of the "C language" handwriting program, as set forth in Schedule C, and to use such source and object code to copy, translate, modify, improve, and enhance said "C" language" handwriting recognition program and to develop derivative works thereof;
>
> 6.1.2 to promote, market, and export said "C language" handwriting recognition program and derivative works thereof; and
>
> 6.1.3 to copy, have copied, have made, distribute, lease, and otherwise transfer and sublicense said "C language" handwriting recognition program and derivative works thereof in source and object code.

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6.1.4 to utilize source code of the "C language" handwriting program pursuant to Section 6.1.3 hereunder; provided, however, that any such transfer or license of source code be conditioned upon the execution of a confidentiality agreement and upon the giving of notice to GRiD 15 days prior to any such transfer of license. Such notice shall identify the party to whom the source code will be transferred, shall specify the source code being so transferred or licensed, and shall include a copy of the confidentiality agreement to be entered into.

The new agreement enabled both Hawkins and GRiD/Tandy to share information licensed to them with sublicensees, provided notice was given and confidentiality agreements were executed.

Though Tandy continued to worry that Palm might somehow use the "C language" enhancements to compete directly with its products, Hawkins' decision to allow Tandy to participate in the first round of equity financing quieted their fears sufficiently to let the deal proceed.

Hawkins saw that the Tandy investment could be very positive for Palm. Tandy's obvious interest in the Palm products made it a very likely first customer. If Tandy committed to carrying the product line, Hawkins might be able to tap into Tandy's well-established network of manufacturing partners—a very valuable entrée since Palm had not yet established a strong performance record of its own.

To ally his fears about Tandy taking over the company at some future date, Hawkins capped their total holdings at 10% and limited their participation in corporate governance to board observer status. Terms were agreed in December 1991. Palm opened for business on January 2, 1992, and the final agreement was executed on January 21, 1992.

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EPILOG

- **Cross-License Negotiated**
  - Hawkins rights to "C" Language
  - GRiD/Tandy rights to sublicense software
- **Tandy investment in Palm 1st round**
  - $300,000
  - Capped at 10% (act. 9.2%) ownership
  - Board observer only
  - Offsets potential competition from Palm
  - Palm channel to RadioShack market
  - Palm access to Tandy manufacturer network

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